PROTOCOL / VERSION 1.0.0

Render rich model output without trusting it

ModelCanvas is a provider-neutral bridge between model or tool output and purpose-built renderers. Validation, resolution, isolation, and fallback remain explicit at every step.

Envelope lifecycle

1	**Validate**	Exact schema, size, URL and file guards
2	**Resolve**	Type, version, MIME, extension and priority
3	**Load**	Lazy renderer module with error isolation
4	**Render**	Controlled component, catalog or sandbox
5	**Trace**	Timing, capability, errors and fallback path

Trust paths

CONTROLLED	DECLARATIVE	OPEN-ENDED
Pre-registered React components receive validated payloads.	An allow-listed catalog turns data-only schemas into UI.	Generated code runs in a sandbox iframe or terminable Worker.

Security defaults

Every model payload is untrusted. JavaScript URLs, dangerous data URLs, invalid MIME/extension pairs, oversized payloads, function-valued chart formatters, unsafe SVG/HTML, and non-allow-listed network origins are rejected before rendering.

Executable artifacts never receive host cookies, local storage, authentication state, parent DOM access, or same-origin privileges. Stop and reset controls replace the runtime context instead of pretending execution ended.

One versioned contract, many renderer families

Field	Purpose	Validation
id	Stable artifact identity	1-160 characters
type	Renderer discriminator	Exact known type or fallback
version	Protocol compatibility	Semantic version
payload	Renderer-specific data	Exact Zod object
presentation	Non-semantic display hints	Bounded values
security	Trust and sandbox policy	Secure defaults
fallback	Safe degraded representation	Text/image/download only

Fallback is a feature

An unknown type or failed renderer does not become an empty box. The host preserves a safe text, Markdown, image, or download representation and records the exact reason in the Protocol Inspector.